Migration Plan: Generated Data Removal from Git Tracking

> Companion to `data_dependency_analysis.md`

Goal

Transition the repo so that **only raw inputs and reference/config files are tracked**. All processed data, intermediate artifacts, and results are generated by the pipeline.

Phase 1: Pre-Migration Verification (Do First)

Before removing anything, verify that every processed file can be regenerated.

```
# 1. Back up current processed files for comparison
cp -r data/processed/ /tmp/processed_backup/

# 2. Delete all processed files
rm data/processed/*.csv data/processed/*.parquet data/processed/*.json data/processed/
*.geojson

# 3. Regenerate everything
python scripts/audit_step1_boundaries.py
python scripts/audit_step2_firehouses.py
python scripts/audit_step3_precincts.py
python scripts/audit_step4_crashes.py
python scripts/demand_modeling.py

# 4. Compare (should be identical or near-identical)
diff /tmp/processed_backup/firehouses_manhattan.csv data/processed/fire-
houses_manhattan.csv
diff /tmp/processed_backup/demand_lambda_hourly.csv data/processed/de-
mand_lambda_hourly.csv
# ... etc for each file
```

If any file differs: investigate why before proceeding.

Phase 2: Update `.gitignore`

Add these lines to `.gitignore`:

```
# Generated intermediates in raw/
data/raw/*.pkl

# All processed data (generated by pipeline)
data/processed/*.json
data/processed/*.geojson

# Manifests (generated by audit)
data/manifests/*.csv
```

Note: `data/processed/*.csv` and `data/processed/*.parquet` are already in `.gitignore`.

Phase 3: Remove Force-Tracked Files from Git

```
# Remove processed files from git index (keeps local copies)
git rm --cached data/processed/firehouses_clean.csv
git rm --cached data/processed/firehouses_manhattan.csv
git rm --cached data/processed/precincts_manhattan.geojson
git rm --cached data/processed/distance_matrix_firehouse_precinct.csv
git rm --cached data/processed/distance_matrix_firehouse_precinct_manhattan.csv
git rm --cached data/processed/demand_lambda_hourly.csv
git rm --cached data/processed/demand_lambda_dow.csv
git rm --cached data/processed/demand_lambda_precinct.csv
git rm --cached data/processed/demand_model_summary.json

# Remove pkl intermediates from raw/
git rm --cached data/raw/manhattan_geom.pkl
git rm --cached data/raw/cbd_geom.pkl

# Remove manifest
git rm --cached data/manifests/data_audit_summary.csv

# Commit
git commit -m "chore: remove generated data from git tracking

All processed data files are now generated by the pipeline scripts.
See docs/data_dependency_analysis.md for the full generation DAG.
Raw input files and reference boundaries remain tracked."
```

Phase 4: Create Data Download Documentation

Create `data/raw/README.md`:

```
# Raw Data Sources

## Tracked in Repository (no action needed)
- `FDNY_Firehouse_Listing_20260223.csv` — FDNY firehouse locations
- `manhattan_boundary.geojson` — Manhattan borough boundary
- `cbd_boundary.geojson` — CBD/MTA Congestion Relief Zone boundary
- `nyc_borough_boundaries.geojson` — All NYC borough boundaries
- `*_Data_Dictionary.xlsx` — Field definitions for each dataset

## Must Download Manually

### Motor Vehicle Collisions (536 MB)
- **URL**: https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95
- **Expected filename**: `Motor_Vehicle_Collisions_-_Crashes_20260223.csv`
- **Expected rows**: ~2.24 million
- **Verification**: `wc -l` should return approximately 2,240,000

### Police Precincts (3.6 MB)
- **URL**: https://data.cityofnewyork.us/Public-Safety/Police-Precincts/kmub-pusk
- **Expected filename**: `Police_Precincts_20260223.csv`
- **Expected rows**: 78
- **Verification**: `wc -l` should return 79 (header + 78 precincts)
```

Phase 5: Create Pipeline Orchestrator

Create `scripts/run_pipeline.py` that runs the full generation chain:

```
"""End-to-end data processing and experiment pipeline.

Usage:
    python scripts/run_pipeline.py              # full pipeline
    python scripts/run_pipeline.py --step data  # data processing only
    python scripts/run_pipeline.py --step sim   # simulation only (requires data)
"""
```

Key steps to implement:
1. Verify raw data exists (fail fast with download instructions)
2. Run Tier 1-3 data processing
3. Run optimization
4. Run verification/validation
5. Run production experiments

Phase 6: Fix Makefile

Replace the broken `data` target:

```
data:
	@echo "Step 1: Processing boundaries..."
	python scripts/audit_step1_boundaries.py
	@echo "Step 2: Processing firehouses..."
	python scripts/audit_step2_firehouses.py
	@echo "Step 3: Processing precincts..."
	python scripts/audit_step3_precincts.py
	@echo "Step 4: Processing crashes..."
	python scripts/audit_step4_crashes.py
	@echo "Step 5: Demand modeling..."
	python scripts/demand_modeling.py
	@echo "Data processing complete."
```

Rollback Plan

If anything goes wrong, the removed files are still in git history:

```
# Restore any specific file from the commit before removal
git checkout HEAD~1 -- data/processed/firehouses_manhattan.csv
```

Checklist

- [ ] Phase 1: Verify all processed files can be regenerated
- [ ] Phase 2: Update `.gitignore` with new patterns
- [ ] Phase 3: `git rm --cached` all force-tracked generated files
- [ ] Phase 4: Create `data/raw/README.md` with download instructions
- [ ] Phase 5: Create pipeline orchestrator script
- [ ] Phase 6: Fix Makefile data target
- [ ] Final: Run full pipeline on a clean checkout to validate